

I'm (not) just a statistician

Working in multidisciplinary research as an ECR

Jiadong Mao

Melbourne Integrative Genomics

School of Mathematics and Statistics

University of Melbourne

Stats department in 2019 (=1994+25)



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Machine Learning, 45, 5–32, 2001

Random Forests

LEO BREIMAN

Statistics Department, University of California, Berkeley, CA 94720

Statistical Science

2001, Vol. 16, No. 3, 199–231

Statistical Modeling: The Two Cultures

Leo Breiman

Berkeley Statistics

About

People

Academics

Research

Industry

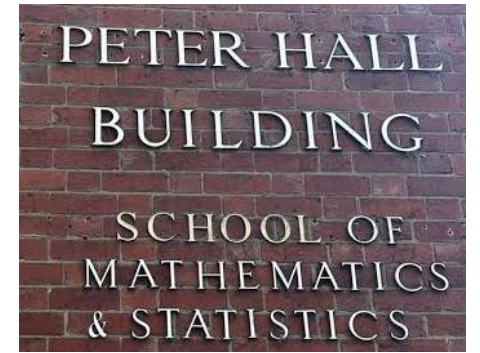
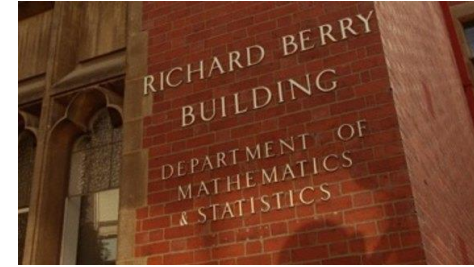
Leo Breiman's Commencement Address, 1994

What is the Statistics Department 25 Years From Now?

As you wander from room to room listening at each open door (yes the doors are now wide open and inviting, and the halls cozy), in one room people are looking at Fourier transforms of speech waves pinned up on walls and discussing the grammatical structure of the English sentence. In this room, three MDs and two young statisticians are working over the details of a ten year study of brain cancer treatments. In another, some astronomers and Professor Stark are arguing about how strongly the data points to evidence of a Big Bang origin for the universe.

How I ended up in bioinformatics

- 2014–2020: MSc & PhD from School of Maths & Stats (MS), UniMelb (Aurore Delaigle)
- 2020–2022: first postdoc at MS, mostly in lockdowns
- Biggest lessons from lockdown: connection with people; scientific problems; a bit of immunology
- Since July 2022: Melbourne Integrative Genomics (MIG) & MS



Renaming of MS building



Lockdown residence
Next to General Cemetery



Aurore Delaigle



Kim-Anh Lê Cao



Melbourne Integrative Genomics (MIG)



Omics and modern statistics

- Bulk RNA sequencing, eg microarray
- High dimension, low sample size
 - ~50 samples, >10,000 genes
- Variable selection, multiple testing
- Eg breast cancer subtyping, eg HER2, Triple-Negative

J. R. Statist. Soc. B (2005)
67, Part 3, pp. 427–444

Geometric representation of high dimension, low sample size data

Peter Hall,
Australian National University, Canberra, Australia
J. S. Marron
University of North Carolina, Chapel Hill, USA
and Amnon Neeman
Australian National University, Canberra, Australia

JOURNAL ARTICLE

The Group Lasso for Logistic Regression FREE

Lukas Meier ✉, Sara Van De Geer, Peter Bühlmann

Journal of the Royal Statistical Society Series B: Statistical Methodology, Volume 70, Issue 1, February 2008, Pages 53–71, <https://doi.org/10.1111/j.1467-9868.2007.00627.x>

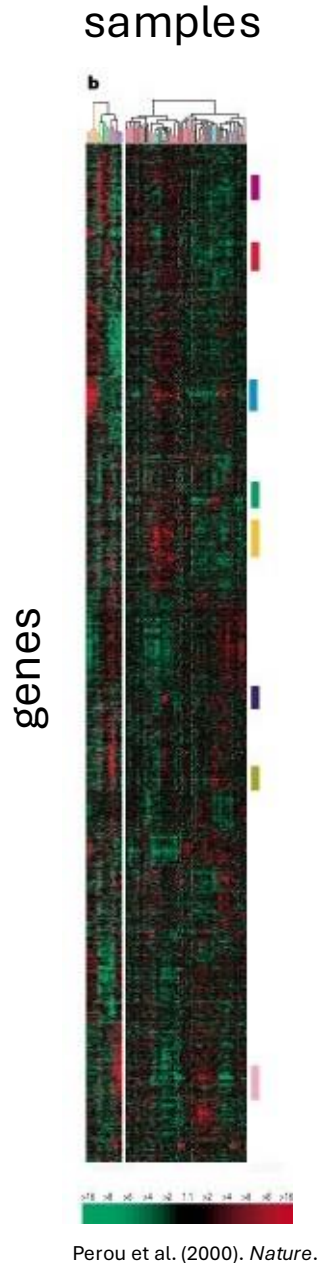
Published: 04 January 2008 Article history ▼

The Annals of Applied Statistics
2007, Vol. 1, No. 1, 107–129
DOI: 10.1214/07-AOAS101
© Institute of Mathematical Statistics, 2007

ON TESTING THE SIGNIFICANCE OF SETS OF GENES

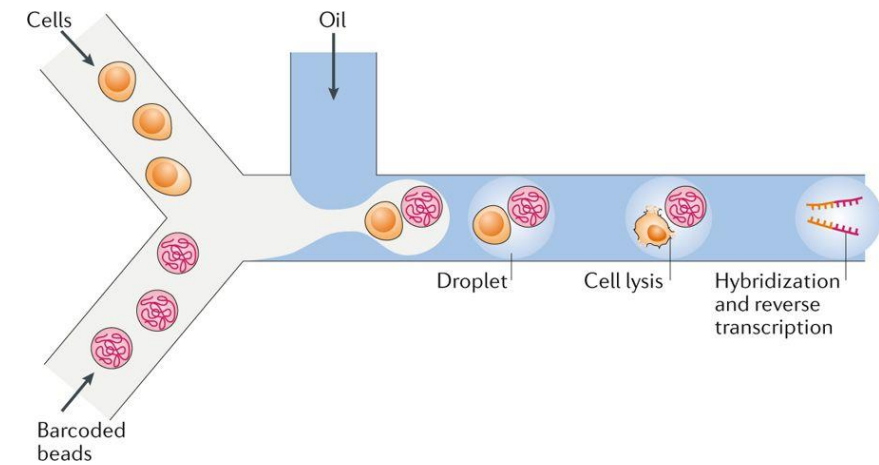
BY BRADLEY EFRON¹ AND ROBERT TIBSHIRANI²

Stanford University

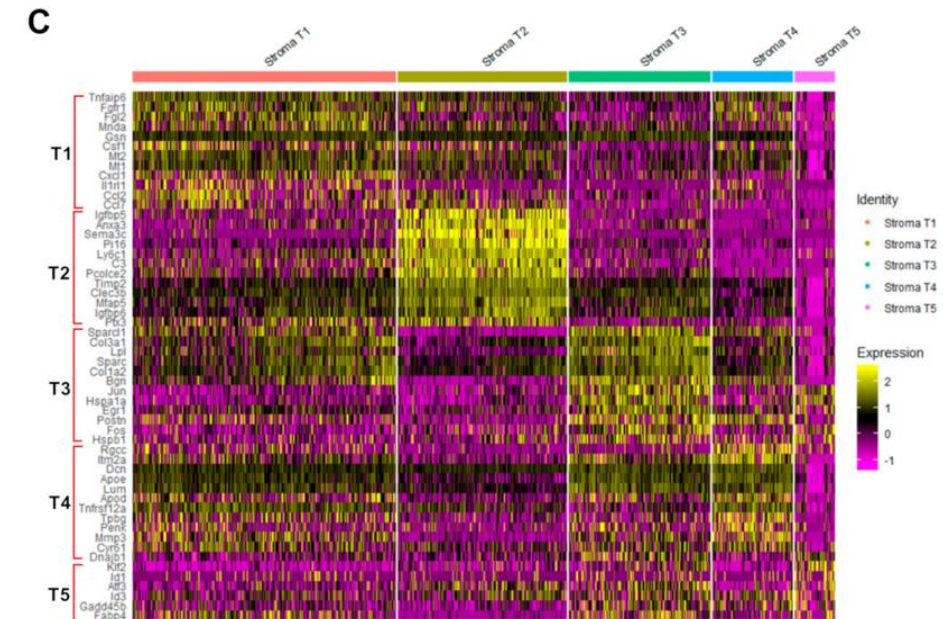


New challenges: big omics data

- Single-cell RNA sequencing (scRNA-seq)
 - Tissue level averages → single cells
- High dimension, large sample size
 - >10,000 cells per sample/donor
 - >20,000 genes
- Key step: finding **cell types**

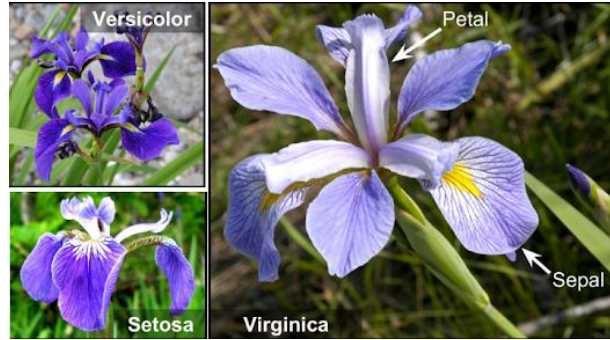


<https://www.rna-seqblog.com/wp-content/uploads/2018/08/droplet.png>

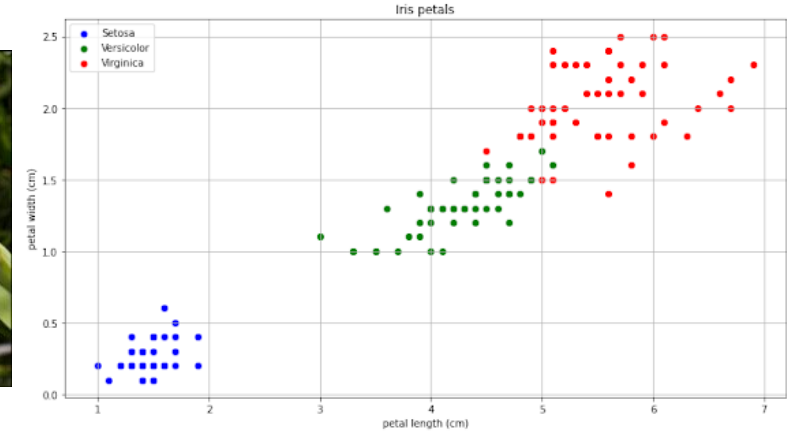


Isn't it just classification?

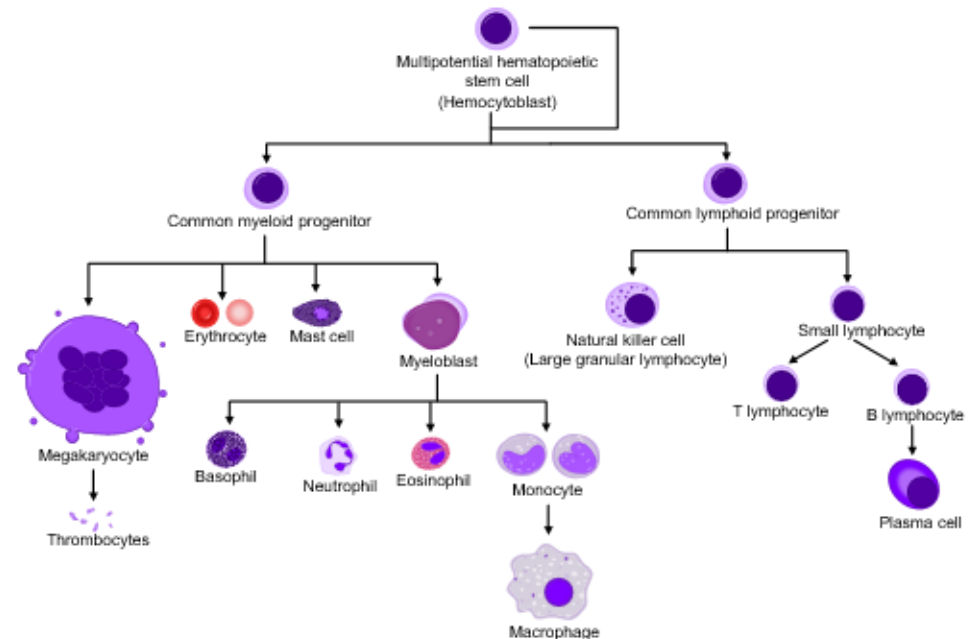
- One cell, one label
- But:
 - Batch effect
 - Normalisation
 - Ambiguity of 'cell type'
- What is a 'cell type'
 - Cluster of cells with similar gene expression
 - Interpret clusters via genes
 - But cells are changing
 - Discrete cell types more common in blood
 - Less common in e.g. bone marrow



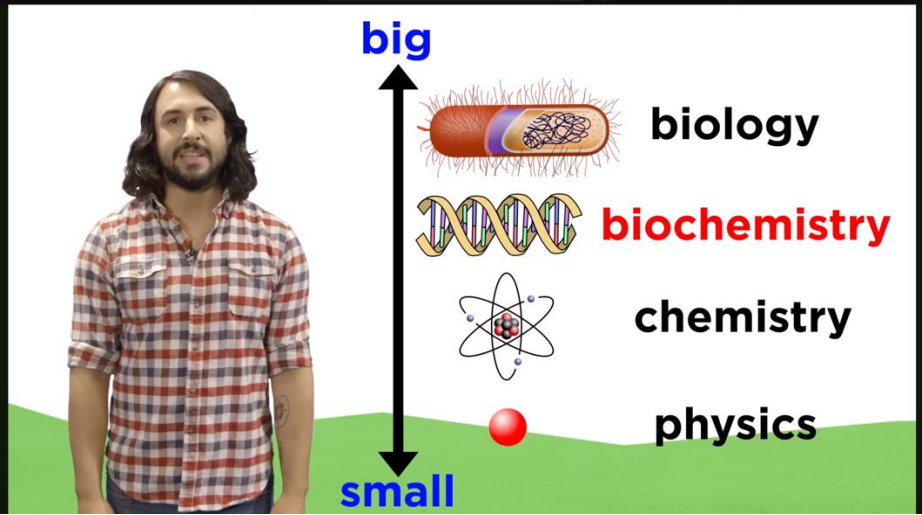
Fisher's 1936 LDA paper,
Annals of Eugenics



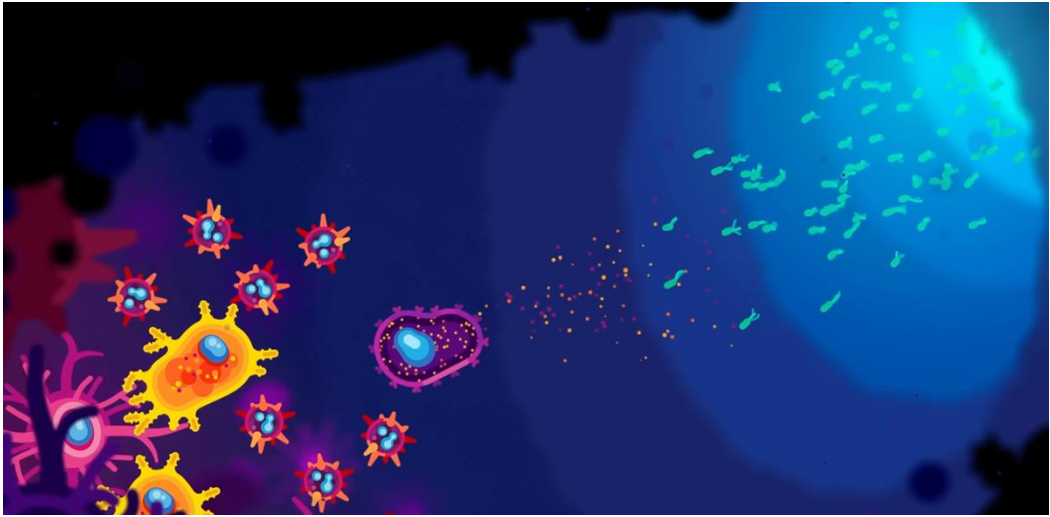
<https://www.quarkml.com/2022/05/iris-dataset-classification-with-python.html>



Re-education



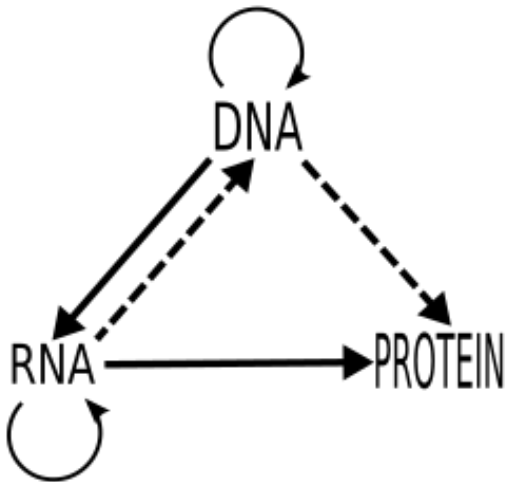
YouTube: Prof Dave Explains



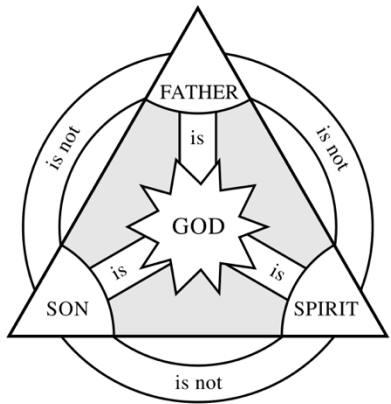
YouTube: Kurzgesagt – In a Nutshell



Most useful of all: talk to biologists!

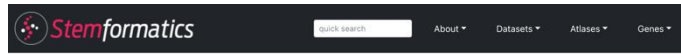


Crick's central dogma



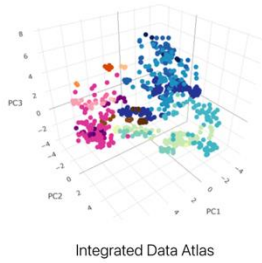
Another central dogma

Learn from fellow scientists



Welcome to Stemformatics

Visualise hundreds of gene expression datasets



Christine Wells (MDHS)



Jarny Choi (St Vincent)



Kim-Anh Lê Cao (MIG)



Yidi Deng (ANU & Florey)



Mark Dawson (Roche)



Dane Vassiliadis



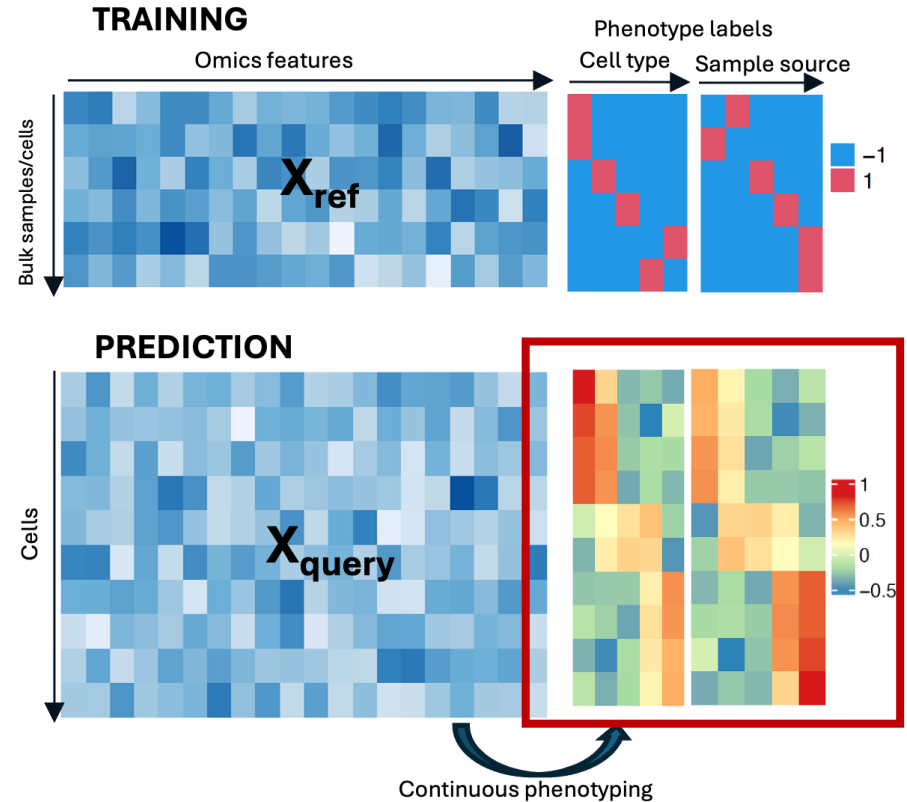
Sarah-Maria Fendt



Xiao-Zheng (Teeny) Liu

From problem to method

- Continuous prediction
- Stats intuition: prediction as dimension reduction



**The Elements of
Statistical Learning**
Data Mining, Inference, and Prediction

Journal of the
Royal Statistical Society

J. R. Statist. Soc. B (2012)
74, Part 2, pp. 267–286

SERIES B
Statistical
Methodology



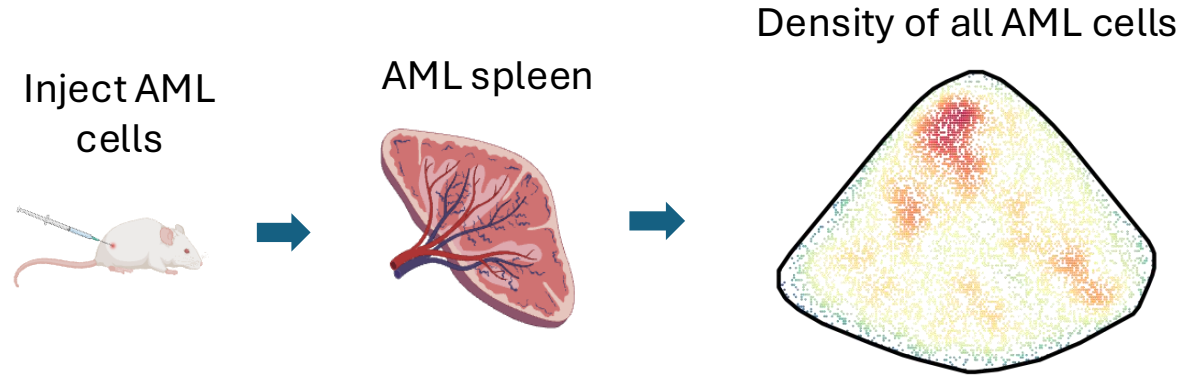
**Achieving near perfect classification for functional
data**

Aurore Delaigle and Peter Hall
University of Melbourne, Australia

Ex. 4.3 Suppose we transform the original predictors $\mathbf{X}$ to $\hat{\mathbf{Y}}$ via linear regression. In detail, let $\hat{\mathbf{Y}} = \mathbf{X}(\mathbf{X}^T\mathbf{X})^{-1}\mathbf{X}^T\mathbf{Y} = \mathbf{X}\hat{\mathbf{B}}$, where $\mathbf{Y}$ is the indicator response matrix. Similarly for any input $x \in \mathbb{R}^p$, we get a transformed vector $\hat{y} = \hat{\mathbf{B}}^T x \in \mathbb{R}^K$. Show that LDA using $\hat{\mathbf{Y}}$ is identical to LDA in the original space.

From problem to method

- PhiSpace
 - Soft classification
 - Downstream analysis of scores
- This turns out to be useful for stem cells & **spatial omics**
- Eg cell states in acute myeloid leukaemia (M Dawson Lab, Peter Mac)



Mao et al. *Genome Biology* (2025) 26:323
<https://doi.org/10.1186/s13059-025-03755-8>

Genome Biology

METHODOLOGY

Open Access

Φ -Space: continuous phenotyping of single-cell multi-omics data

Jiadong Mao¹, Yidi Deng^{1,2} and Kim-Anh Lê Cao^{1,4,*}

Cell Reports Methods

CellPress
OPEN ACCESS

Article

Φ -Space ST: A platform-agnostic method to identify cell states in spatial transcriptomics studies

Jiadong Mao^{1,2}, Jarry Choi^{2,3} and Kim-Anh Lê Cao^{1,4,*}

¹Melbourne Integrative Genomics, School of Mathematics and Statistics, The University of Melbourne, Parkville, VIC 3010, Australia

²Bioinformatics and Cellular Genomics, St Vincent's Institute of Medical Research, Fitzroy, VIC 3065, Australia

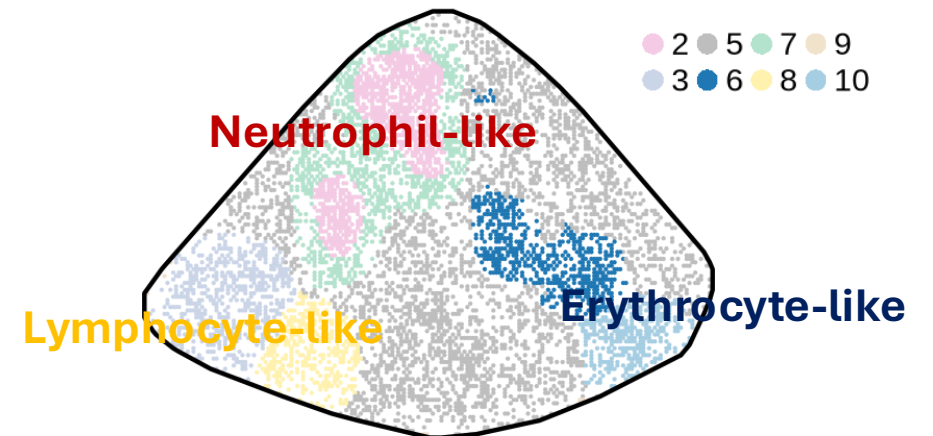
³These authors contributed equally

⁴Lead contact

*Correspondence: kimanh.lecao@unimelb.edu.au

<https://doi.org/10.1016/j.crmeth.2026.101483>

AML meta-clones based on PhiSpace



Connecting people

- Statisticians can play in backyards of immunologists, clinicians and biologists
- Knowledge sharing

- Φ -Space for doublet detection and composition analysis



Michelle Meier
(Peter Mac)



George Howitt
(Peter Mac)



Alicia Oshlack
(Peter Mac)

- Longitudinal patient stratification: MD Anderson
→ UQ + Flinders; Wisconsin-Madison



Kim-Anh Do
(MD Anderson)



Ranjeny Thomas
(UQ)



Mihir Wechalekar
(Flinders)



Kris Sankaran
(Wisconsin-Madison)

- StableMate, DIVAS: early-onset breast cancer



Shuai Li
(MSPGH)



Agus Salim
(M&S, MSPGH)

Landmark \$15 million boost for
early-onset cancer research

 Cancer Australia
Research Initiative

Lessons?

- Multilingualism
- Curiosity about the question
- Real problems generate methods



(This is not AI)



(This is not AI either)